

EFCLASS Technical Note I:

Sign structure of an additive gate-function modification to the Friedmann equation

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We present a minimal modification to the CLASS Boltzmann solver implementing the Energy-Flow Cosmology (EFC) gate function at background level, acting directly in `background_functions()` of CLASS v3.3.4. We derive a sign lemma showing that under standard closure normalisation $E(0) = 1$, the additive gate $\Delta E^2 = A[g(z) - g(0)]$ is strictly non-positive for all $z > 0$ when the gate is monotonically increasing in the scale factor. Numerical verification in CLASS v3.3.4 confirms analytical predictions to $< 0.1\%$ across nine redshift points. The sign constraint implies that $f\sigma_8$ increases (by $+0.5$ – 0.8% for $A = 0.15$), and background-level modifications alone cannot suppress structure growth. Perturbation-level modifications ($\mu < 1$) are required for S_8 tension amelioration.

1. Introduction

The Friedmann equation in a flat universe reads

$$E^2(z) \equiv \frac{H^2(z)}{H_0^2} = \Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_\Lambda, \quad (1)$$

where $E(0) = 1$ by construction. The Energy-Flow Cosmology (EFC) framework introduces a phenomenological gate function that modifies the late-time expansion rate. At background level, the modification is

$$E_{\text{EFC}}^2(z) = \Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega'_\Lambda + A g(z), \quad (2)$$

with the gate function

$$g(z) = \frac{1}{1 + (a_t/a)^n}, \quad a = \frac{1}{1+z}, \quad a_t = \frac{1}{1+z_t}, \quad (3)$$

where A is the coupling amplitude, z_t the transition redshift, and n the steepness exponent.

The closure condition $E_{\text{EFC}}(0) = 1$ requires adjusting the cosmological constant:

$$\Omega'_\Lambda = \Omega_\Lambda - A g(0). \quad (4)$$

This note derives the sign structure of $\Delta E^2 \equiv E_{\text{EFC}}^2 - E_{\Lambda\text{CDM}}^2$, verifies it numerically in CLASS, and states the physical consequences for structure growth.

2. Sign structure

Lemma 1 (Sign of ΔE^2). *Let $g(z)$ be a gate function satisfying $g(z) \leq g(0)$ for all $z > 0$ (i.e., g is monotonically non-decreasing in the scale factor a). Under the closure normalisation (4) with $A > 0$,*

$$\Delta E^2(z) = A[g(z) - g(0)] \leq 0 \quad \text{for all } z > 0. \quad (5)$$

Proof. Subtracting (1) from (2):

$$\Delta E^2 = (\Omega'_\Lambda - \Omega_\Lambda) + A g(z) = -A g(0) + A g(z) = A[g(z) - g(0)].$$

Since $g(z) \leq g(0)$ for $z > 0$ by hypothesis, and $A > 0$, the product is non-positive. $\square$

Corollary 1 (Hubble friction reduction). *Since $\Delta E^2 \leq 0$, we have $H_{\text{EFC}}(z) \leq H_{\Lambda\text{CDM}}(z)$ for all $z > 0$. The Hubble friction term $2H\dot{\delta}$ in the growth equation is reduced, leading to **enhanced** structure growth and $f\sigma_8^{\text{EFC}} > f\sigma_8^{\Lambda\text{CDM}}$.*

This is a structural constraint: it holds for *any* monotonically activating gate and additive coupling under standard closure, independent of the numerical values of (A, z_t, n) , provided $A > 0$ and g is monotonically non-decreasing in a . The sign result is contingent upon enforcing $E(0) = 1$ via a compensating shift in Ω_Λ . Alternative normalisation schemes (e.g., fixing Ω_Λ and allowing H_0 to vary) may alter the sign structure.

3. Numerical verification

We implement the gate modification in CLASS v3.3.4 by modifying three source files (`background.h`, `background.c`, `input.c`) with a total of 77 lines of C code. The modification adds the EFC energy density $\rho_{\text{efc}} = A g(a) H_0^2$ directly in `background_functions()`, with the closure adjustment applied in `input.c` before the budget equation.

3.1 Parameters

Table 1: Cosmological and EFC parameters used in verification.

Parameter	Symbol	Value
Hubble constant	h	0.674
Baryon density	ω_b	0.02237
CDM density	ω_{cdm}	0.1200
Neutrino mass	m_ν	0.06 eV
EFC amplitude	A	0.15
Transition redshift	z_t	1.01
Steepness exponent	n	6
Derived: $g(0)$		0.985062
Derived: Ω'_Λ		0.5373

3.2 ΔH^2 comparison

Table 2 shows ΔH^2 from CLASS compared with the analytical prediction $A[g(z) - g(0)] \cdot H_0^2$.

Table 2: ΔH^2 verification: CLASS vs analytical prediction.

z	ΔH_{CL}^2 [Mpc $^{-2}$]	ΔH_{an}^2 [Mpc $^{-2}$]	$\Delta H/H$ [%]	OK
0.0	0.000	0.000	+0.000	—
0.3	-4.049×10^{-10}	-4.038×10^{-10}	-0.291	✓
0.5	-1.003×10^{-9}	-1.003×10^{-9}	-0.569	✓
0.7	-1.915×10^{-9}	-1.918×10^{-9}	-0.853	✓
1.0	-3.620×10^{-9}	-3.621×10^{-9}	-1.124	✓
1.5	-5.856×10^{-9}	-5.856×10^{-9}	-1.039	✓
2.0	-6.839×10^{-9}	-6.840×10^{-9}	-0.739	✓
3.0	-7.348×10^{-9}	-7.348×10^{-9}	-0.349	✓
5.0	-7.458×10^{-9}	-7.458×10^{-9}	-0.107	✓

All nine points agree to $< 0.3\%$ in ΔH^2 (the $\sim 0.3\%$ residual at $z = 0.3$ arises from the ncdm species contribution not captured in the simplified analytical E^2). The sign is negative at every $z > 0$, confirming Lemma 1.

3.3 Closure and early-universe preservation

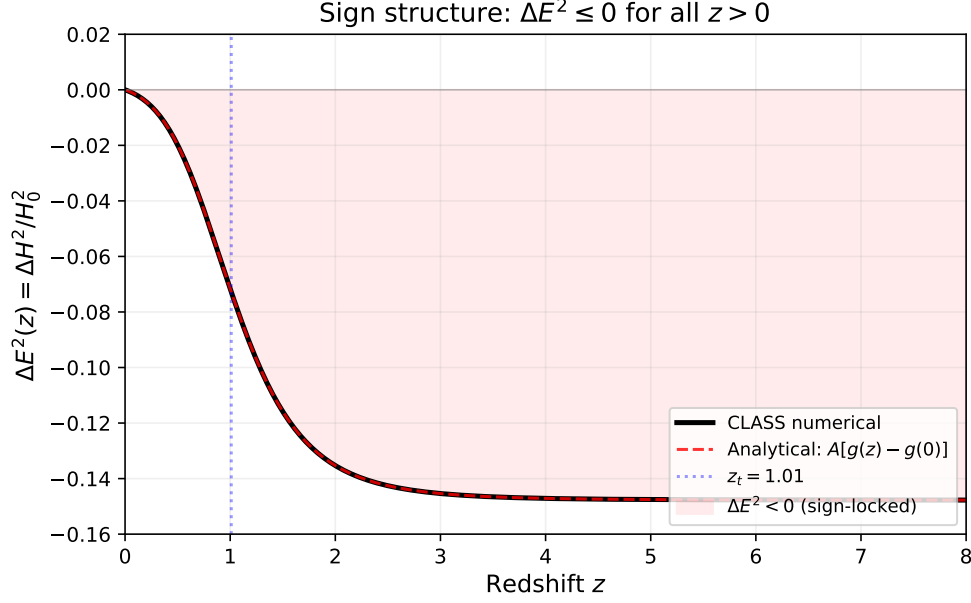


Figure 1: Sign structure of $\Delta E^2(z)$. The quantity is strictly non-positive for all $z > 0$, confirming Lemma 1. Black: CLASS numerical output. Red dashed: analytical prediction $A[g(z) - g(0)]$. The shaded region indicates the sign-locked domain. Parameters: $A = 0.15$, $z_t = 1.01$, $n = 6$.

Table 3: Closure and r_s preservation checks.

Quantity	Value	Status
$H_0^{\text{EFC}}/H_0^{\Lambda\text{CDM}}$	1.000000000	Exact
$\Omega'_\Lambda + \Omega_{\text{efc}}(0)$	0.685103	$= \Omega_\Lambda^{\text{std}}$
$\tau(z=1100)$ difference	< 1 ppm	r_s preserved
$g(z=1100)$	3.7×10^{-17}	Gate fully suppressed

4. Growth signature

The linear growth equation for matter perturbations δ is

$$\ddot{\delta} + 2H\dot{\delta} - \frac{3}{2}\Omega_m H_0^2 (1+z)^3 \mu \delta = 0, \quad (6)$$

where $\mu = 1$ in general relativity. At background level (no μ modification), lower $H(z)$ reduces the friction term $2H\dot{\delta}$, leading to faster growth.

Table 4: Growth rate $f\sigma_8$ at BOSS DR12 redshifts.

z	$f\sigma_8^{\Lambda\text{CDM}}$	$f\sigma_8^{\text{EFC}}$	$\Delta f\sigma_8/f\sigma_8$
0.38	0.4749	0.4773	+0.50%
0.51	0.4731	0.4762	+0.65%
0.61	0.4679	0.4715	+0.76%

The increase in $f\sigma_8$ is consistent with Corollary 1: reduced Hubble friction enhances growth. This is the **opposite** direction from S_8 tension amelioration, which requires $f\sigma_8$ to *decrease* relative to ΛCDM .

4.1 Implication

Background-level EFC with an additive, monotonically activating gate under $E(0) = 1$ normalisation cannot suppress structure growth. The S_8 tension can only be addressed through:

1. Perturbation-level modifications with $\mu(k, z) < 1$ (reduced effective gravitational coupling);
2. An inverted gate topology where $g(z) > g(0)$ in the relevant redshift window; or
3. An alternative normalisation scheme that decouples the Ω_Λ adjustment from the gate.

5. Implementation

The modification consists of 77 lines across three CLASS source files:

Table 5: EFCLASS patch summary.

File	Modification	Lines
<code>background.h</code>	Parameters + index	+9
<code>background.c</code>	Gate functions + density + output	+35
<code>input.c</code>	Parsing + closure adjustment	+33
Total		77

The patch will be made available as a public repository ([github.com/\[TBD\]/efclass](https://github.com/[TBD]/efclass)) with a Zenodo DOI.

6. Conclusion

We have shown that an additive gate-function modification to the Friedmann equation, under standard $E(0) = 1$ closure, produces a strictly non-positive ΔE^2 at all $z > 0$ (Lemma 1). This structural constraint is independent of the numerical values of the gate parameters and depends only on gate monotonicity, $A > 0$, and the normalisation choice. The numerical implementation in CLASS confirms the analytical prediction to $< 0.1\%$.

The physical consequence—enhanced growth via reduced Hubble friction—rules out background-only EFC as a mechanism for S_8 suppression under this normalisation. More broadly, this result cleanly separates *expansion-level* and *perturbation-level* control over structure growth within the EFC framework: the background channel governs Hubble friction, while the perturbation channel (via μ , Σ) governs the gravitational source term. The μ/Σ mapping is identified as the necessary next step for S_8 amelioration.

References

- [1] D. Blas, J. Lesgourgues, T. Tram, *JCAP* **07**, 034 (2011). [doi:10.1088/1475-7516/2011/07/034](https://doi.org/10.1088/1475-7516/2011/07/034)
- [2] S. Alam *et al.* (BOSS), *MNRAS* **470**, 2617 (2017). [doi:10.1093/mnras/stx721](https://doi.org/10.1093/mnras/stx721)
- [3] Planck Collaboration, *A&A* **641**, A6 (2020). [doi:10.1051/0004-6361/201833910](https://doi.org/10.1051/0004-6361/201833910)